AP CALCULUS AB

Unit 1 • Lesson 1.6 practice • Answer key

Lesson 1.6 • Determining Limits Using Algebraic Manipulations

4 multiple-choice questions and 1 free-response question • Answer key with explanations

Question 1

$\lim_{x\to -2} \dfrac{x^2-4}{x+2}$ is

(A) 0
(B) -4
(C) 4
(D) It does not exist.

Solution

Answer: (B). $\dfrac{(x-2)(x+2)}{x+2} = x-2 \to -4.$

Question 2

$\lim_{x\to 4} \dfrac{\sqrt{x}-2}{x-4}$ is

(A) $\dfrac{1}{4}$
(B) $\dfrac{1}{2}$
(C) 0
(D) 4

Solution

Answer: (A). $x-4 = (\sqrt{x}-2)(\sqrt{x}+2)$, so the quotient is $\dfrac{1}{\sqrt{x}+2} \to \dfrac{1}{4}$.

Question 3

$\lim_{h\to 0} \dfrac{(3+h)^2-9}{h}$ is

(A) 9
(B) 0
(C) 6
(D) 3

Solution

Answer: (C). $\dfrac{6h+h^2}{h} = 6+h \to 6.$

Question 4

$\lim_{x\to 1} \dfrac{\frac{1}{x}-1}{x-1}$ is

(A) 1
(B) 0
(C) It does not exist.
(D) -1

Solution

Answer: (D). $\dfrac{\frac{1-x}{x}}{x-1} = -\dfrac{1}{x} \to -1.$

Question 5 (Free response)

Evaluate each limit. Show the algebra.

(a) $\lim_{x\to 5} \dfrac{x^2-3x-10}{x-5}$ (b) $\lim_{x\to 0} \dfrac{\sqrt{4+x}-2}{x}$

(c) $\lim_{x\to 2} \dfrac{x^3-8}{x-2}$ (d) $\lim_{h\to 0} \dfrac{\frac{1}{2+h}-\frac{1}{2}}{h}$

Solution

(a) $\dfrac{(x-5)(x+2)}{x-5} = x+2 \to 7.$

(b) $\dfrac{(4+x)-4}{x(\sqrt{4+x}+2)} = \dfrac{1}{\sqrt{4+x}+2} \to \dfrac{1}{4}.$

(c) $x^3-8 = (x-2)(x^2+2x+4)$, so the limit is $4+4+4=12$.

(d) $\dfrac{\frac{2-(2+h)}{2(2+h)}}{h} = \dfrac{-1}{2(2+h)} \to -\dfrac{1}{4}.$